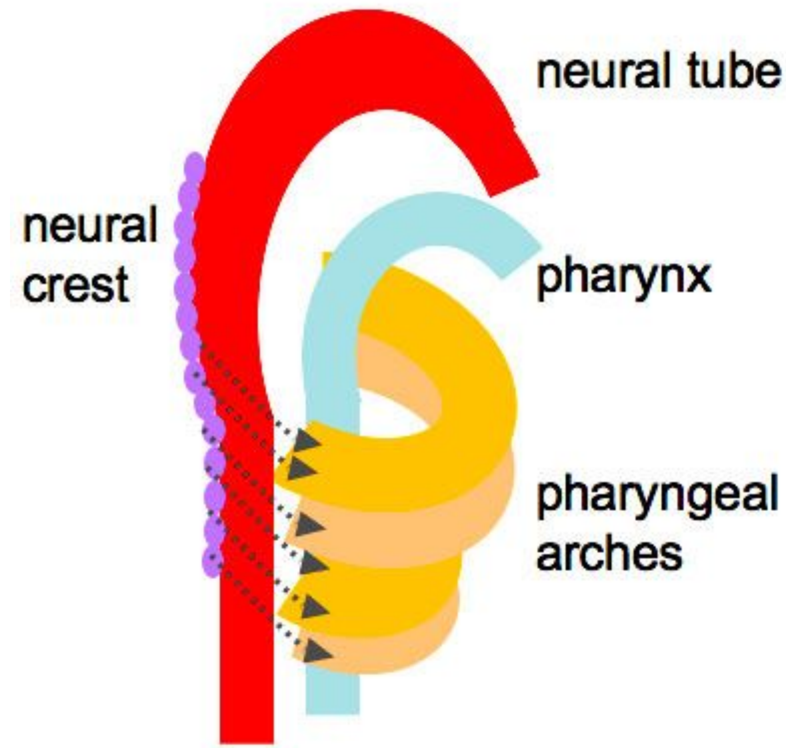
“DERIVATIVES OF PHARYNGEAL ARCHES AND PHARYNGEAL POUCHES (TONGUE,THYROID,THYMUS)”

LEARNING OBJECTIVE

- ***Definition of Branchial Apparatus***
- ***Describe Development of Branchial Apparatus***
- ***Name the different Parts of Branchial Apparatus***
- ***Discuss Fate of Branchial Arch***
- ***Discuss Fate of Branchial Cleft***
- ***Discuss Fate of Branchial Pouch***
- ***Discuss Fate of Branchial Membrane***

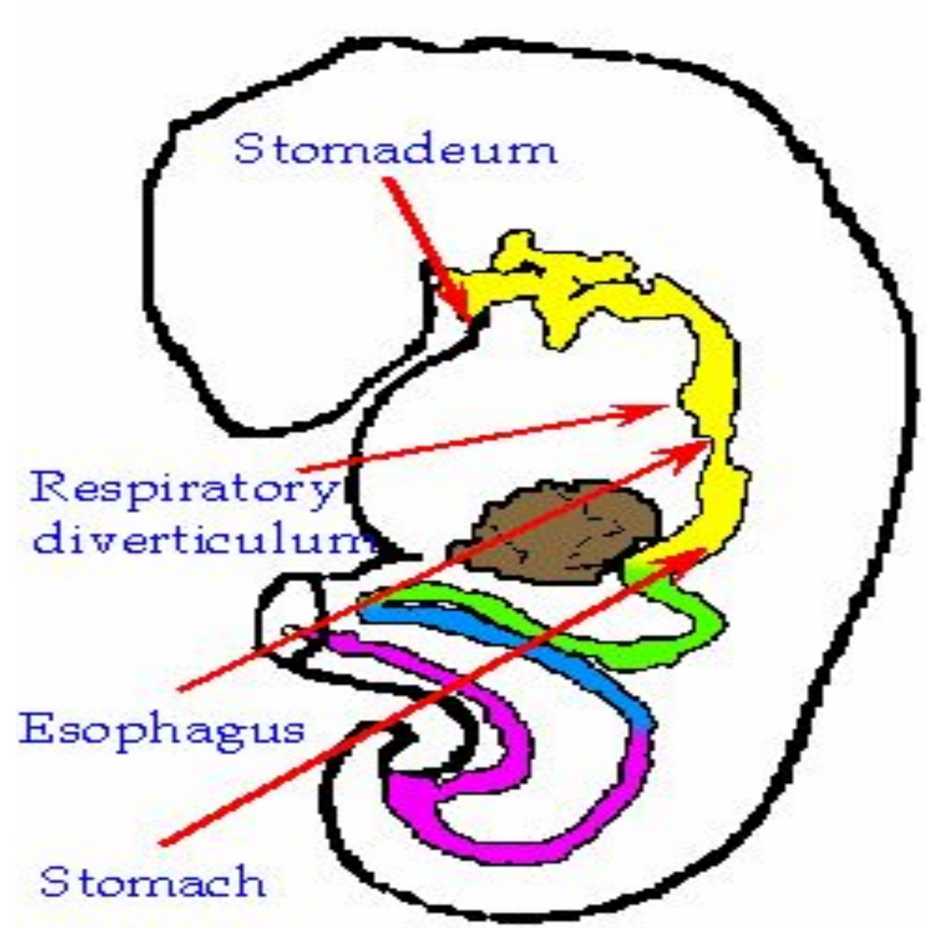
BRANCHIAL APPARATUS

- ***Structure which develops in embryo.***
- ***It is comparable to gills of fish, reflect fact that ontogeny (development of individual) resembles phylogeny (evolution of species).***
- ***But in human the Branchial Apparatus developed into other structures different from gill***



BRANCHIAL APPARATUS

- ***First indication for development of Branchial Apparatus is, at 4th week the Neural Crest cells invade future head and neck region of embryo.***
- ***Cells form ridges on side of head and neck. Located lateral to rostral part of the foregut.***
- ***Will form branchial Apparatus***

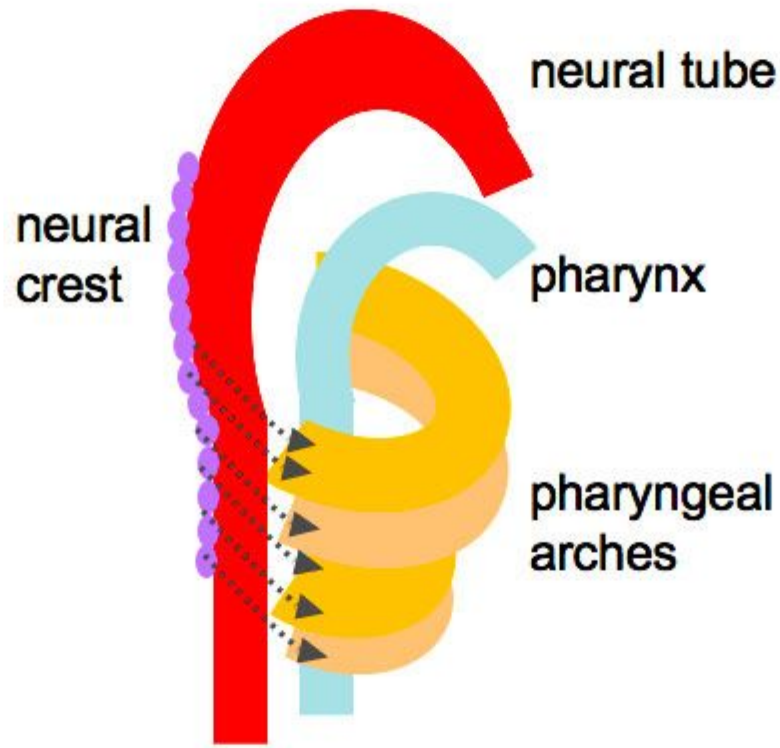


APPEARANCE BRANCHIAL ARCHES

During the fourth week of embryonic development, a series of 5 bar-like ridges appear n the lateral walls of the anterior part of the fore-gut (Pharynx).

These are called as Branchial or Pharyngeal Arches.

These Bar like ridges are formed by the migration of Neural Crest Cells into Cervical Region



COMPONENTS OF BRANCHIAL APPARATUS

There are Brachial Arches which develop in series one above another in the lateral wall of Primitive Pharynx.

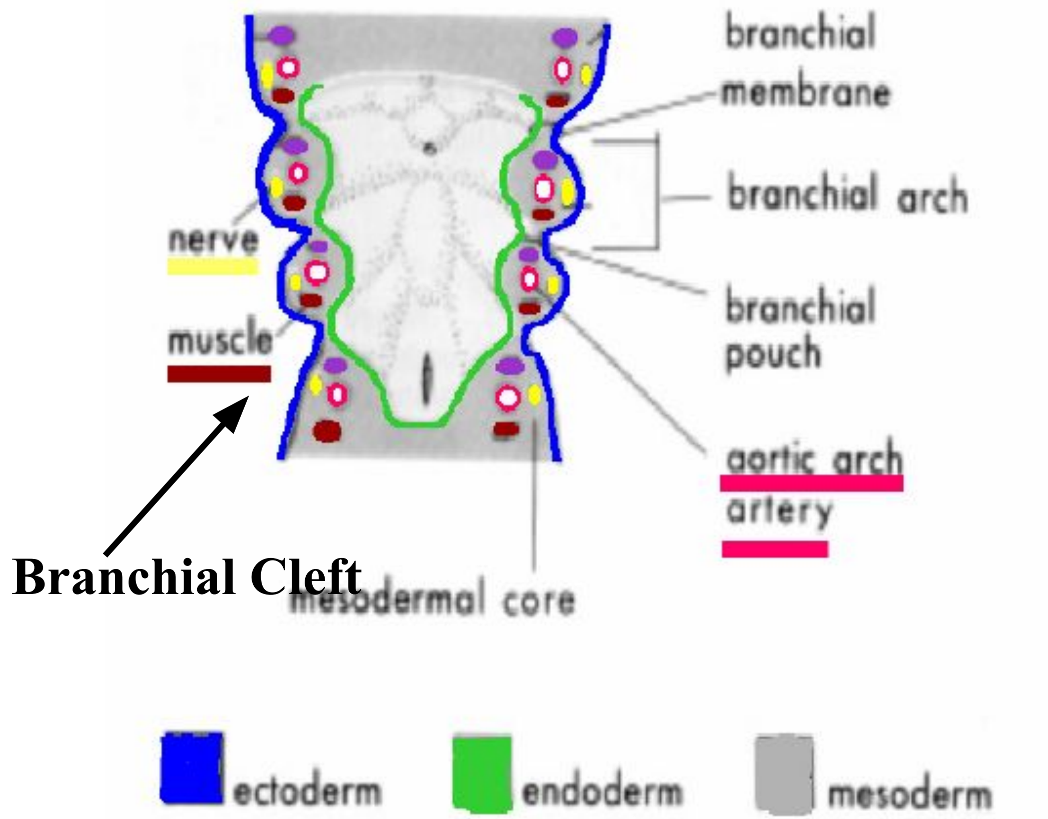
Each Consists of Four Components

Pharyngeal Arch

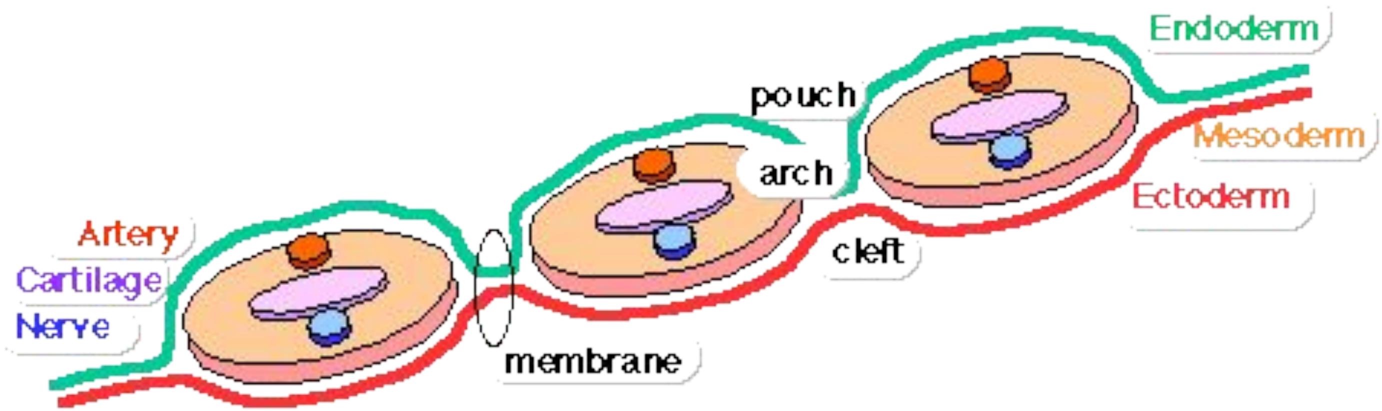
Pharyngeal Pouch

Pharyngeal Groove

Pharyngeal Membrane



PHARYNGEAL ARCH



each arch has initially similar components

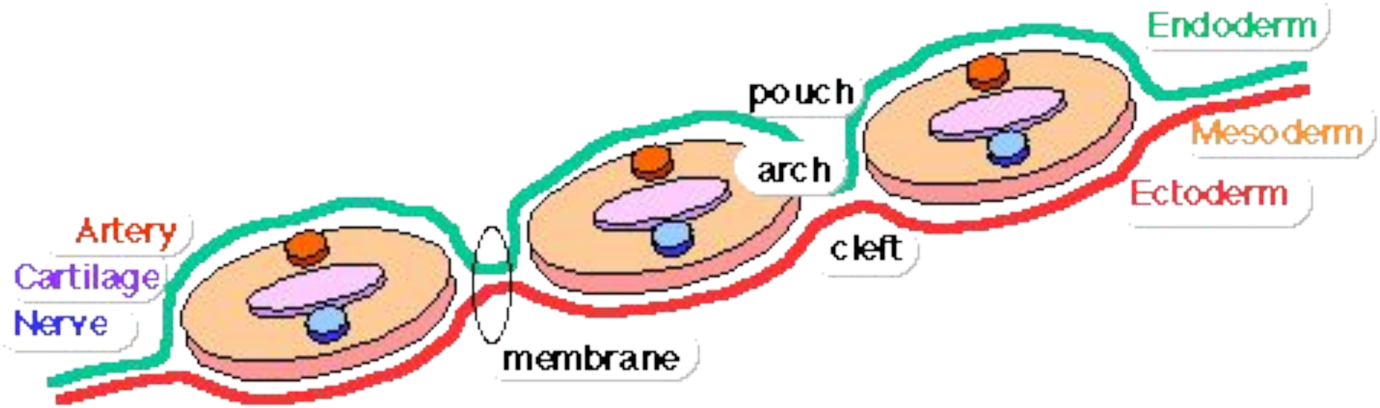
contributions from all 3 germ layers

Ectoderm - outside surface and Neural Crest of core

Mesoderm - core of mesenchyme

Endoderm - inside pharyngeal surface

PHARYNGEAL ARCH FEATURES



Arch

Groove/cleft - externally separates each arch

Pouch - internally separates each arch

Pockets from the pharynx

Membrane - ectoderm and endoderm contact regions

Branchial Apparatus

Structures in Embryonic

Branchial Arches

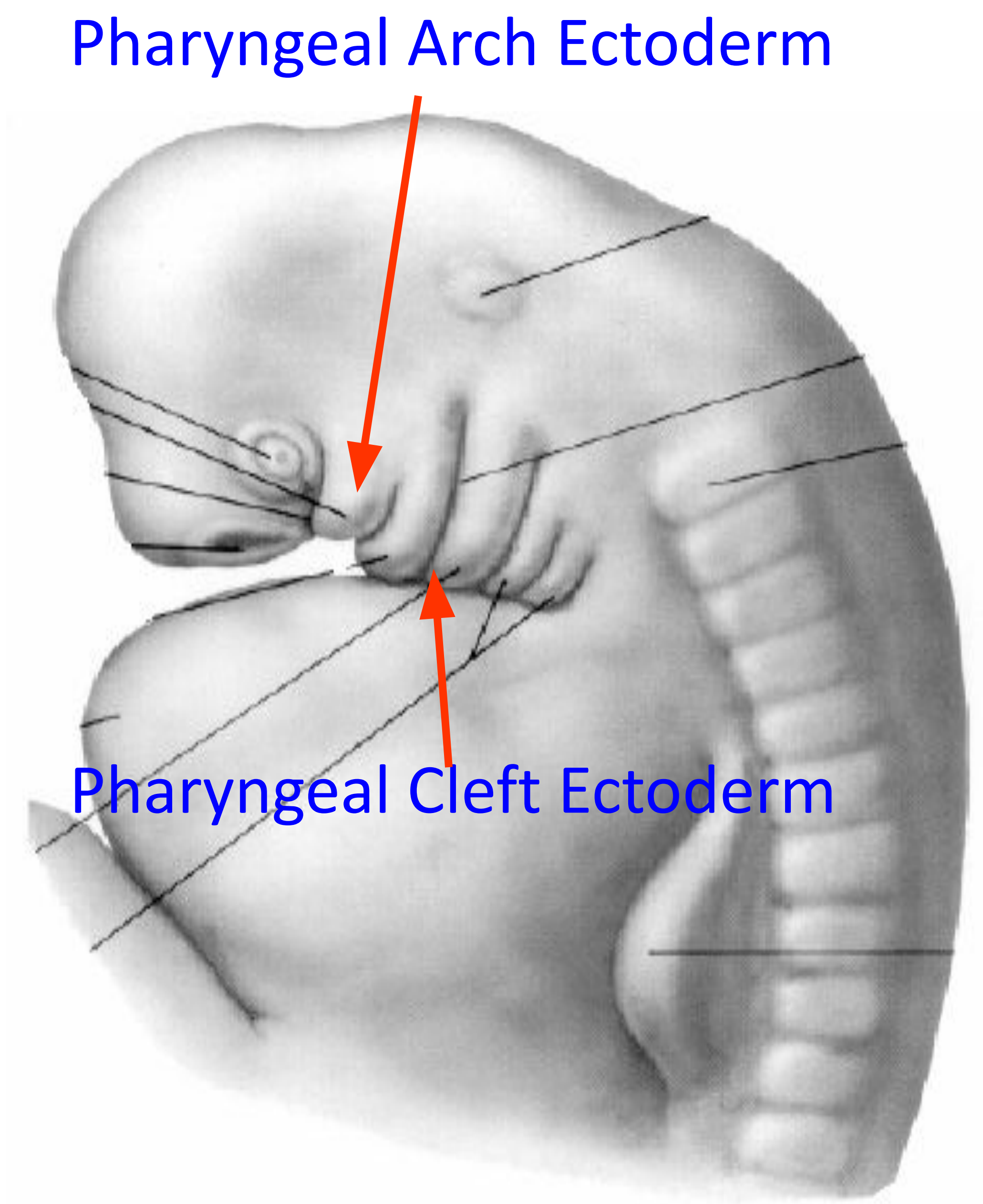
Reorganize to form

cartilages, nerve, muscles

& arteries in fetus. Forms

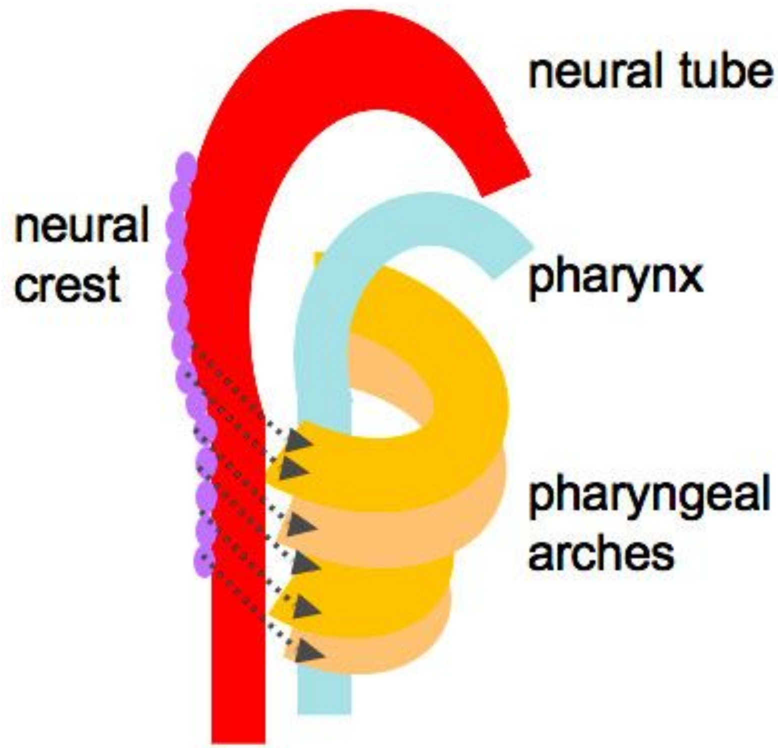
much of musculature

of head some of neck 5- 6 weeks

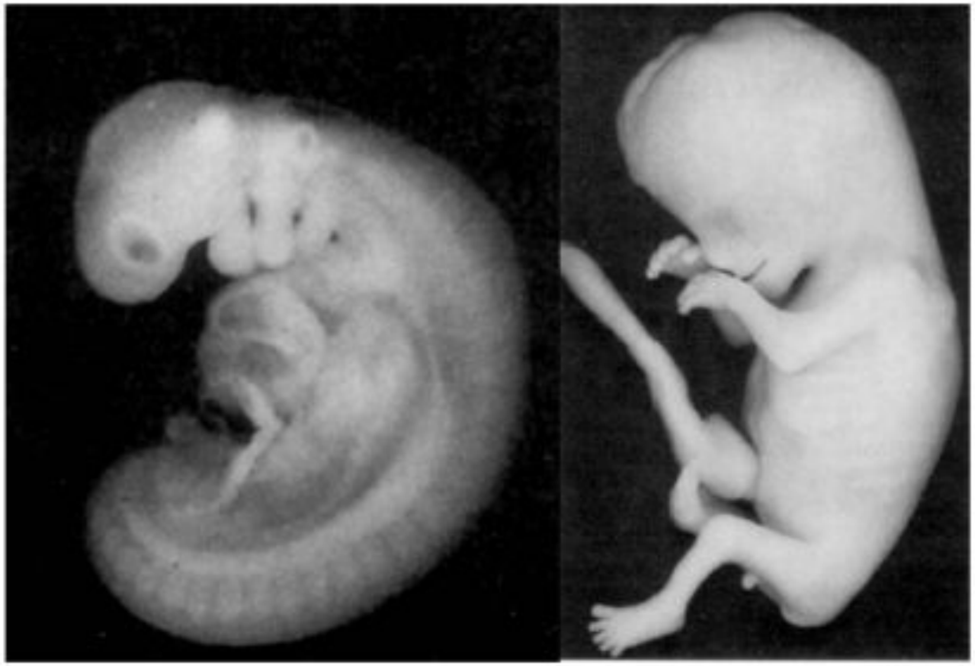


DEVELOPMENT

- ***These grow and join in the ventral midline.***
- ***The first arch, as the first to form, separates the mouth pit or stomodeum from the pericardium.***
- ***By differential growth the neck elongates and new arches form, so the pharynx has six arches ultimately.***



DEVELOPMENT OF BRANCHIAL ARCHES



~4 weeks → ~11 weeks

OUTLINE

I. EARLY DEVELOP/ TERMINOLOGY

II. FATE OF ARCHES (CHART) - CARTILAGES, LIGAMENTS, NERVES, MUSCLES

III. BRANCHIAL POUCHES, GROOVES, MEMBRANES

IV. DEVELOPMENT OF THYROID

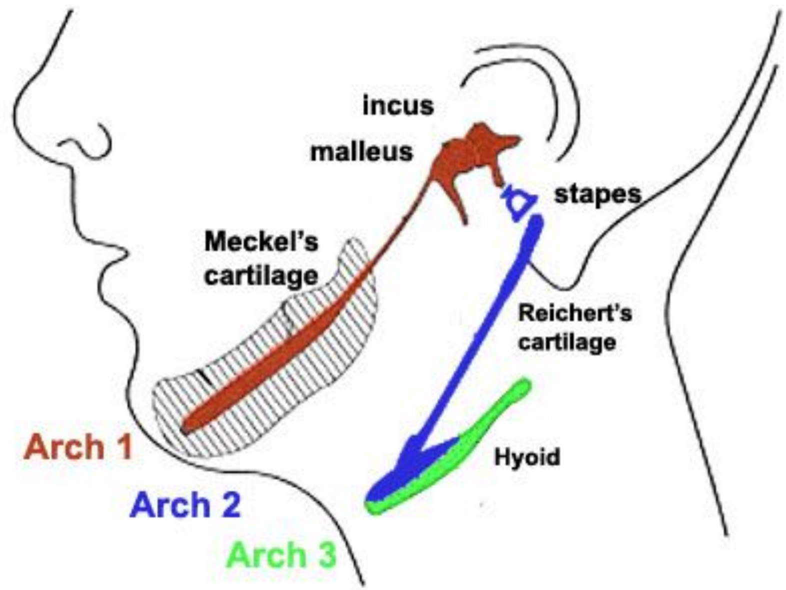
- ADULT STRUCTURE IS RESULT OF TRANSFORMATION;
- SPECIFIC SYNDROMES IF DEVELOPMENT IS INCORRECT

PHARYNGEAL ARCHES

- ***Each pharyngeal arch has a cartilaginous stick.***
- ***A muscle component which differentiates from the cartilaginous tissue.***
- ***An artery.***
- ***A cranial nerve.***
- ***Each of these is surrounded by mesenchyme.***
- ***Arches do not develop simultaneously, but instead possess a "staggered" development***

FATE OF PHARYNGEAL ARCHES

- ***There are six pharyngeal arches.***
- ***But in humans the fifth arch only exists transiently during embryologic growth and development.***
- ***Since no human structures result from the fifth arch.***
- ***Arches in humans are I, II, III, IV, and VI.***



Pharyngeal Arch Structures

RELATIONS

- ***Pharyngeal pouches (or branchial pouches) form on the endodermal side between the arches.***
- ***And pharyngeal grooves (or clefts) form from the lateral ectodermal surface of the neck region to separate the arches.***
- ***The pouches line up with the clefts, and these thin segments become gills in fish.***
- ***In mammals the endoderm and ectoderm not only remain intact, but continue to be separated by a mesoderm layer***

SPECIFIC ARCHES

- ***There are six pharyngeal arches.***
- ***The first three contribute to structures above the larynx, while the last two contribute to the larynx and trachea.***

<u>1ST (CALLED "MANDIBULAR ARCH")</u>

<u>Muscular contributions</u>

Muscles of mastication anterior belly of the digastric, mylohyoid, tensor tympani, tensor veli palatini.

<u>Skeletal contributions</u>

Maxilla,mandible,only as a model for mandible not actual formation of mandible), the incus and malleus of the middle ear, also Meckel's cartilage

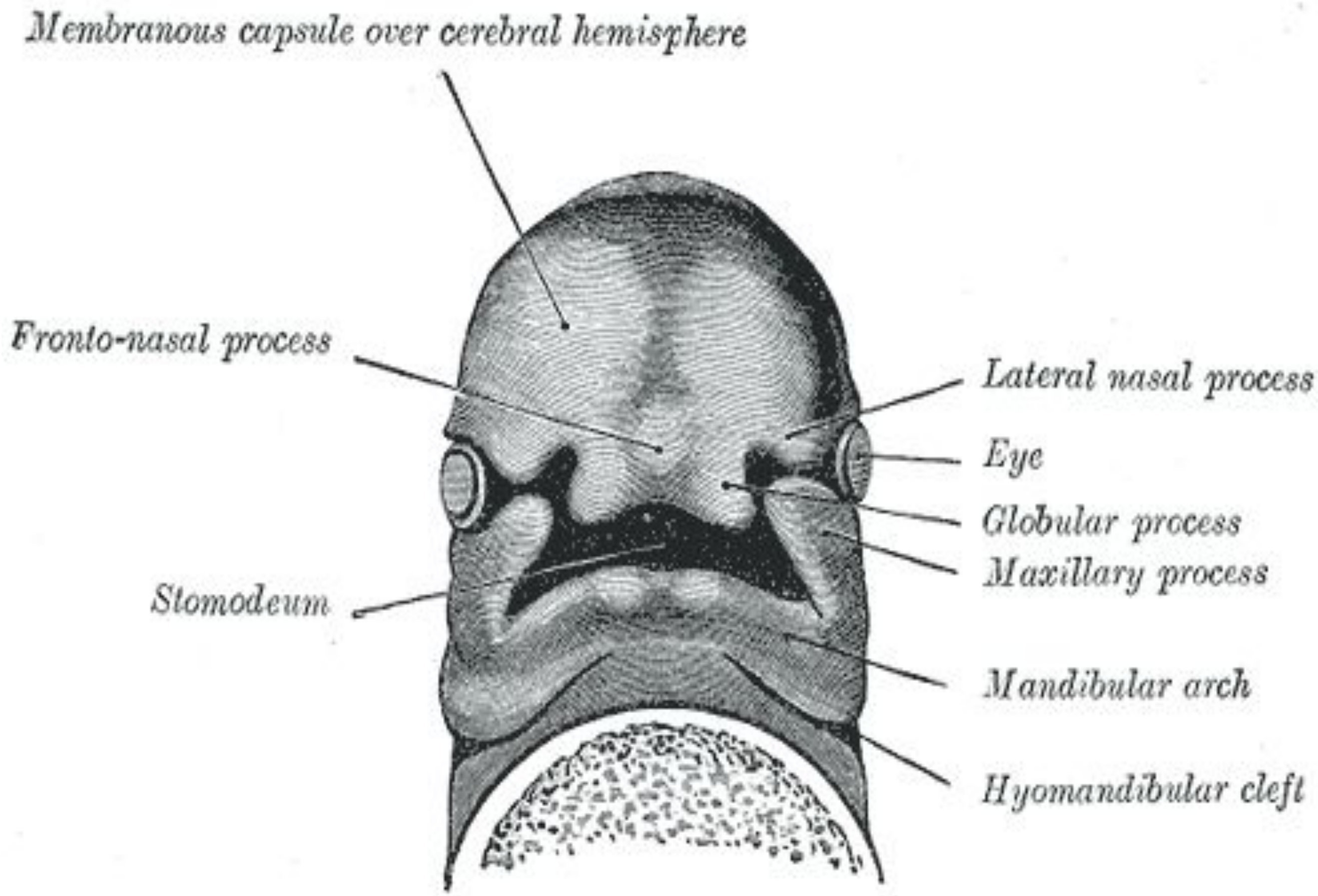
<u>Nerve</u>

Trigeminal nerve.

<u>Artery</u>

Maxillary artery.

1ST (ALSO CALLED "MANDIBULAR ARCH")



2ND (CALLED THE "HYOID ARCH")

Muscular contributions

Muscles of facial expression, buccinator, , platysma, stapedius, stylohyoid, posterior belly of the digastric

Skeletal contributions

Stapes, styloid process, hyoid (lesser horn and upper part of body), Reichert's cartilage

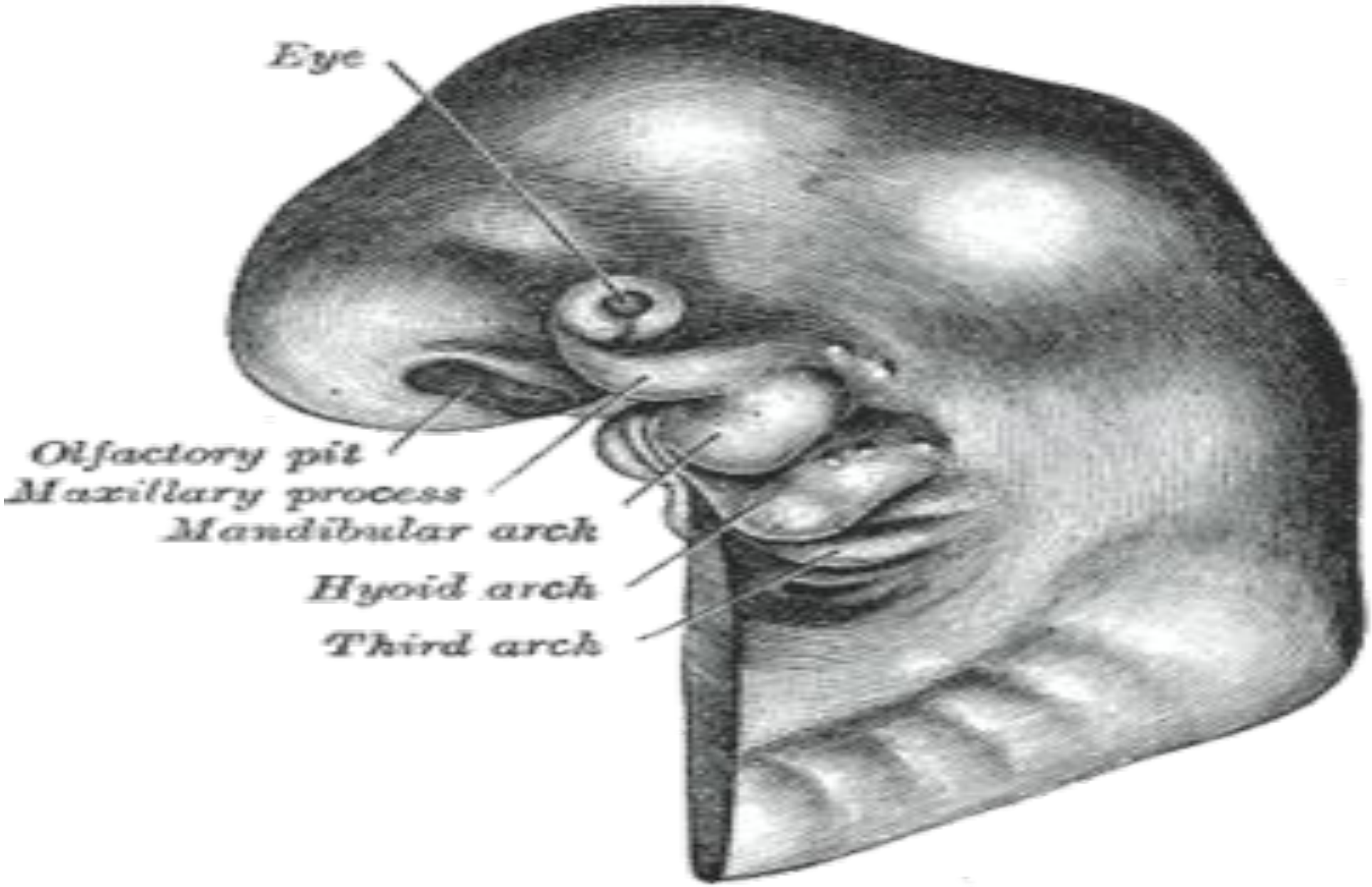
Nerve

Facial nerve (VII)

Artery

Stapedial artery

2ND (CALLED THE "HYOID ARCH")



3RD PHARYNGEAL ARCH

Muscular contributions

Stylopharyngeus

Skeletal contributions

Hyoid (greater horn) and lower part of body), thymus

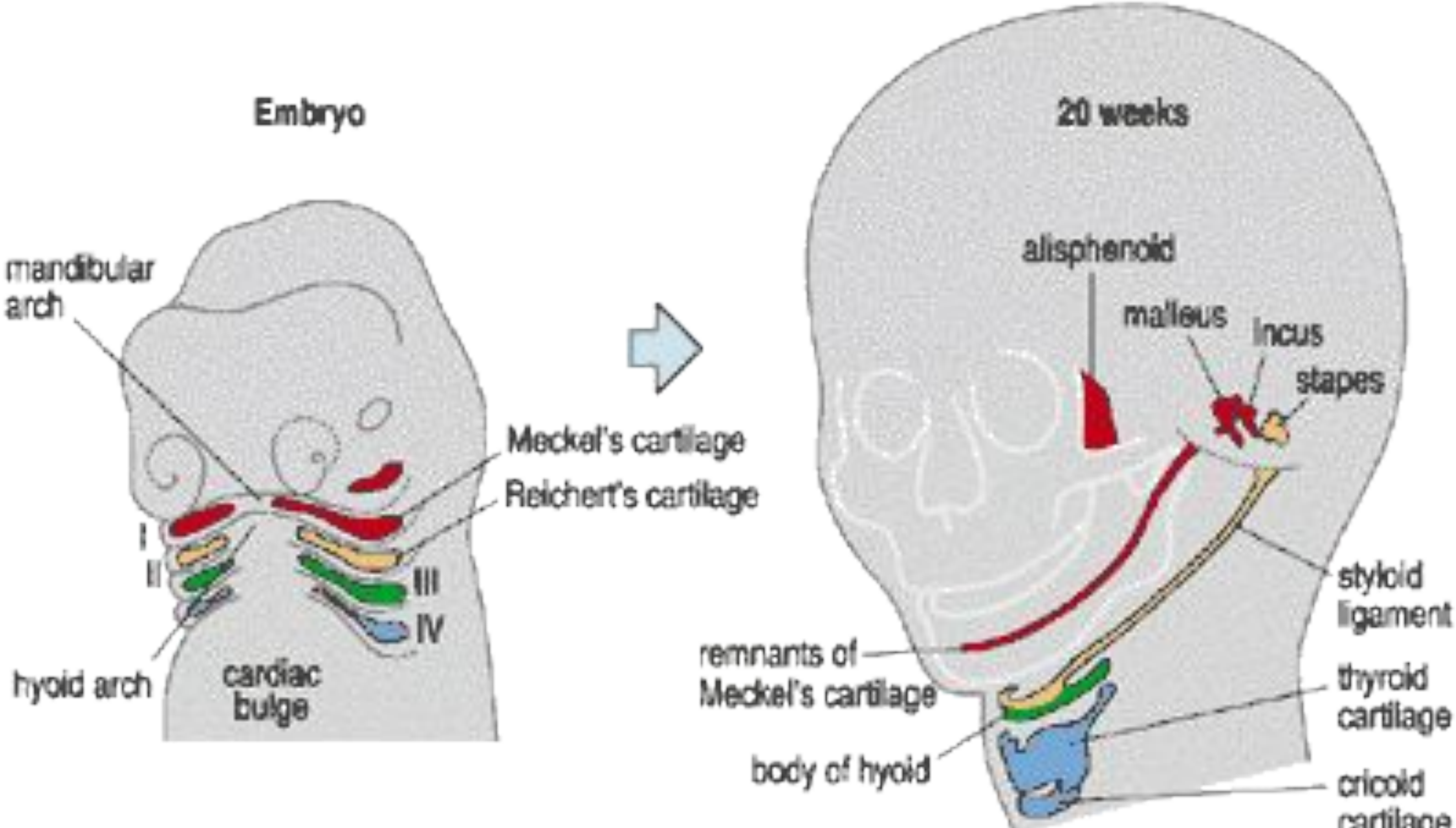
Nerve

Glossopharyngeal nerve (IX)

Artery

Common carotid,internal carotid

3RD PHARYNGEAL ARCH



4^{TH} PHARYNGEAL ARCH

Muscular contributions

Cricothyroid muscle, all intrinsic muscles of of soft palate including levator veli palatini

Skeletal contributions

Thyroid cartilage, epiglottic cartilag0e

Nerve

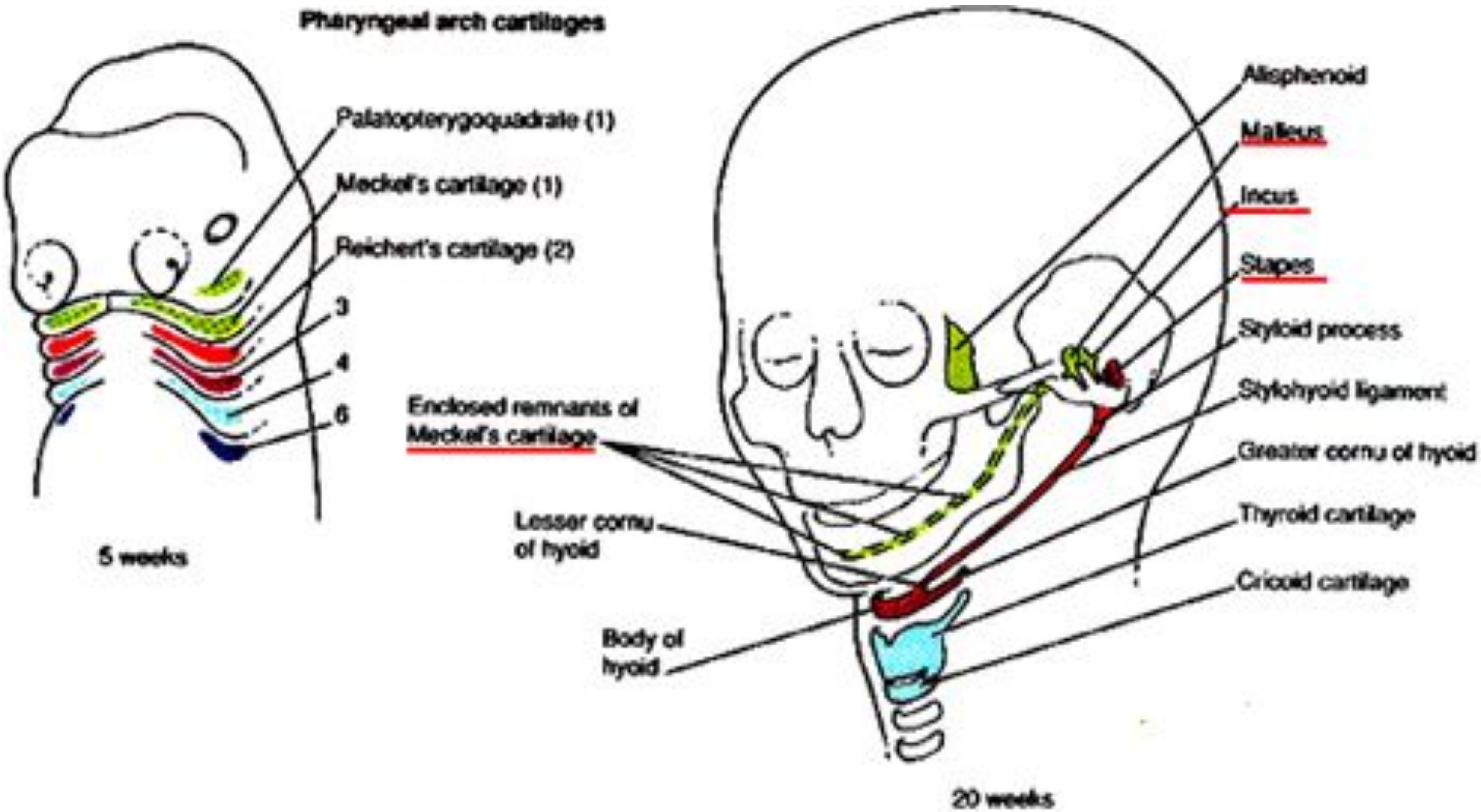
Vagus nerve (X)

Superior laryngeal nerve

Artery

Right 4th aortic arch: subclavian artery Left 4th aortic arch: aortic arch

4^{TH} PHARYNGEAL ARCH



6^{TH} PHARYNGEAL ARCH

Muscular contributions

All intrinsic muscles of larynx except the cricothyroid muscle

Skeletal contributions

Cricoid cartilage,arytenoid cartilage,corniculate cartilage

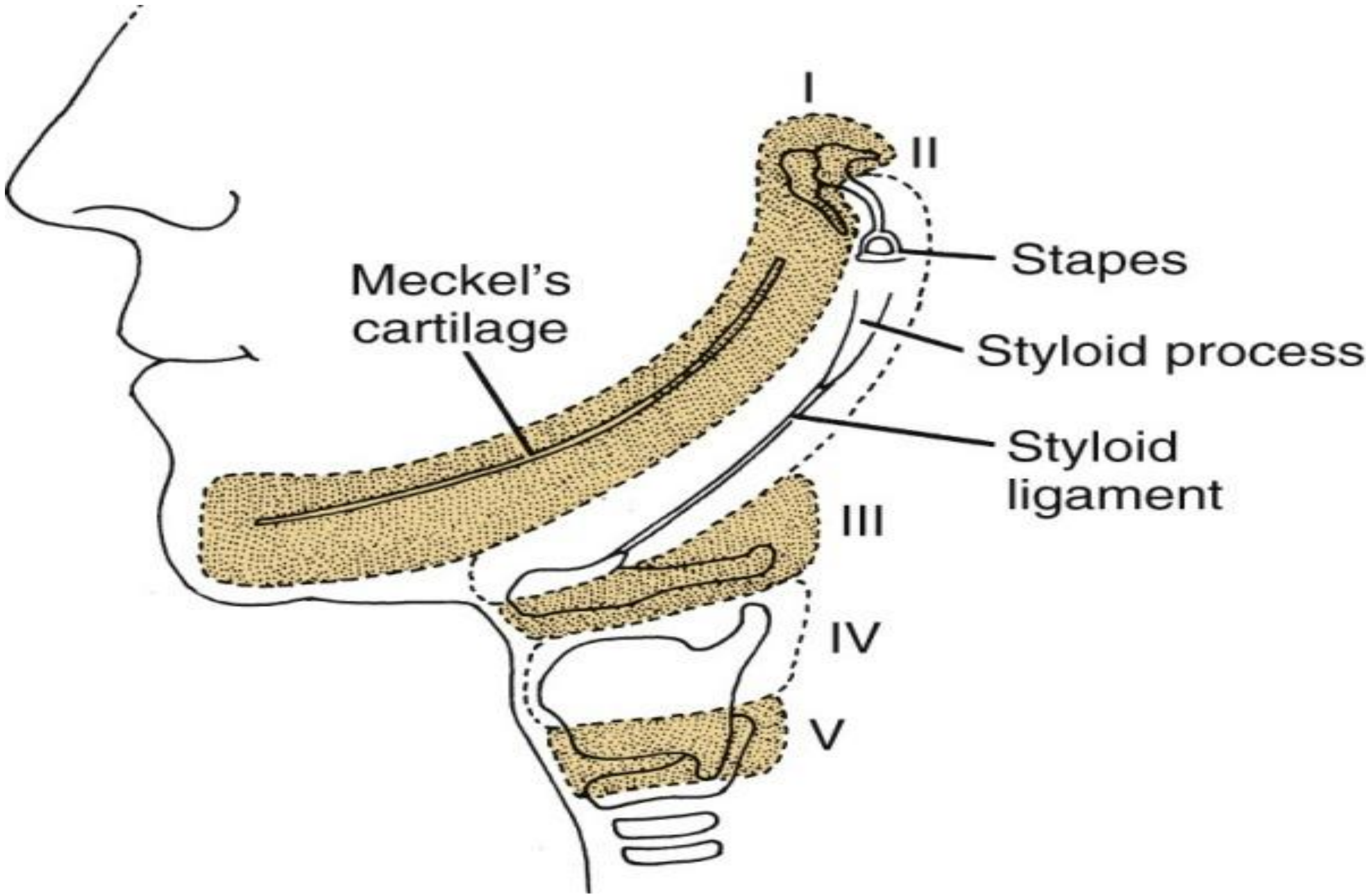
Nerve

Vagus nerve (X) Recurrent laryngeal nerve)

Artery

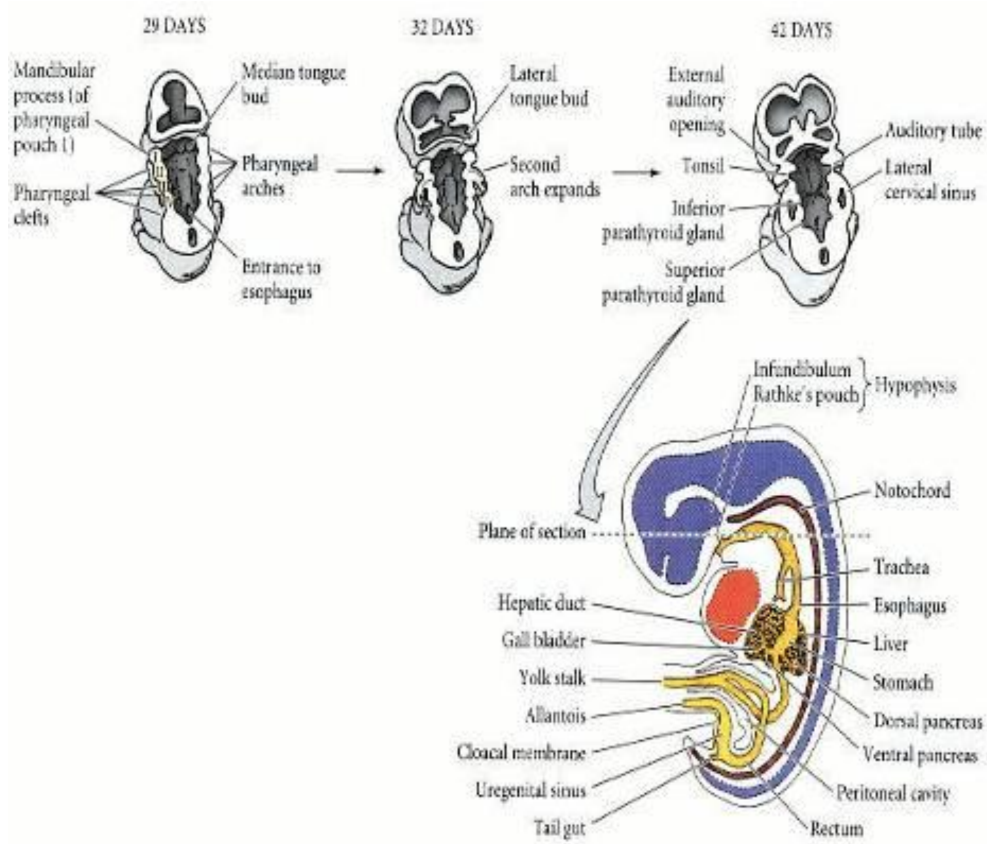
Right 6th aortic arch: pulmonary artery Left 6th aortic arch: Pulmonary artery and ductus arteriosus

6TH PHARYNGEAL ARCH



USE IN STAGING

The development of the pharyngeal arches provide a useful morphological landmark with which to establish the precise stage of embryonic development.



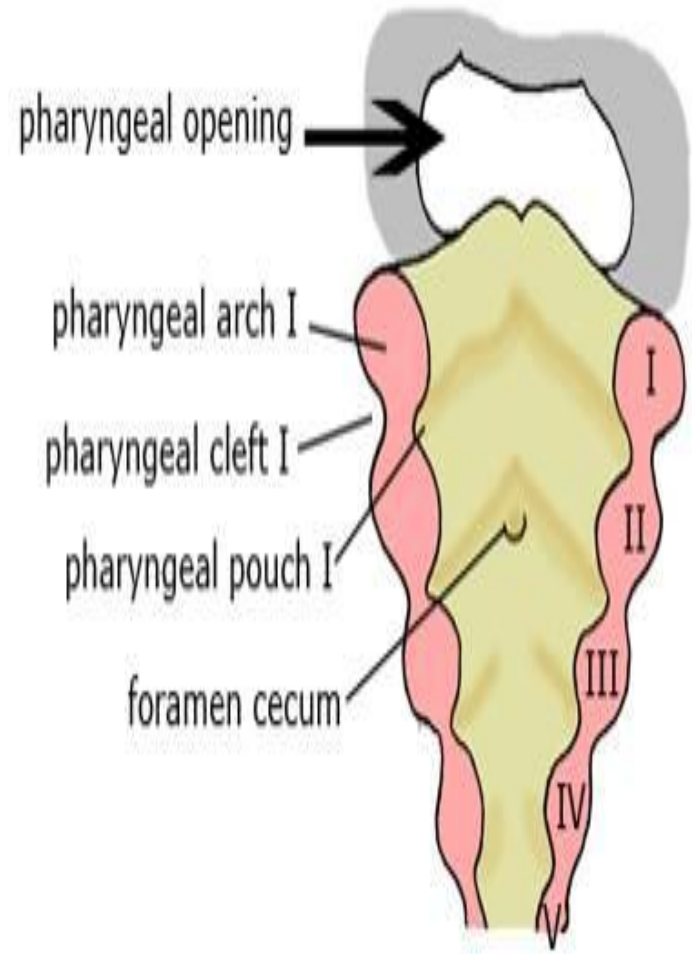
PHARYNGEAL *OR* ***BRANCHIAL POUCHES***

- ***In the development of vertebrate animals, pharyngeal or branchial pouches form on the endodermal side between the branchial arches.***
- ***pharyngeal grooves (or clefts) form the lateral ectodermal surface of the neck region to separate the arches***

FIRST POUCH

The endoderm lines the

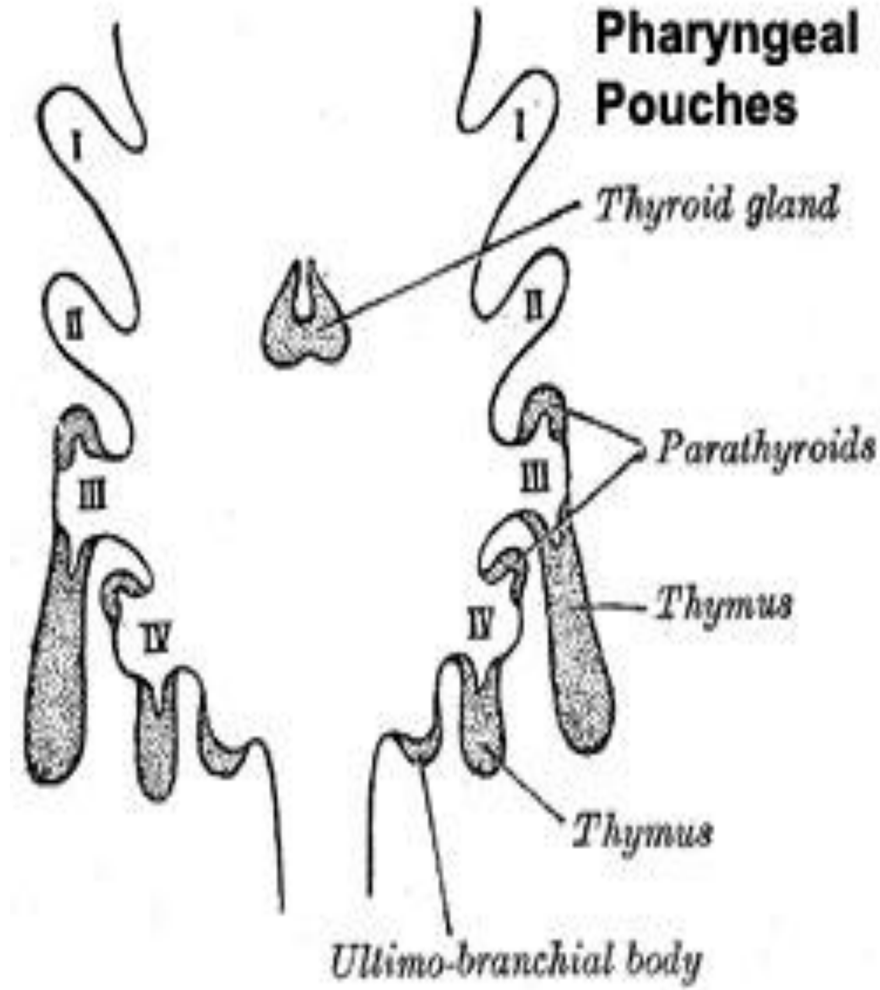
- *future auditory tube (Pharyngotympanic " Eustachian " tube),*
- *middle ear,*
- *mastoid antrum,*
- *and inner layer of the tympanic membrane*



SECOND POUCH

Contributes to the

- *middle ear,*
- *palatine tonsils, supplied by the facial nerve*



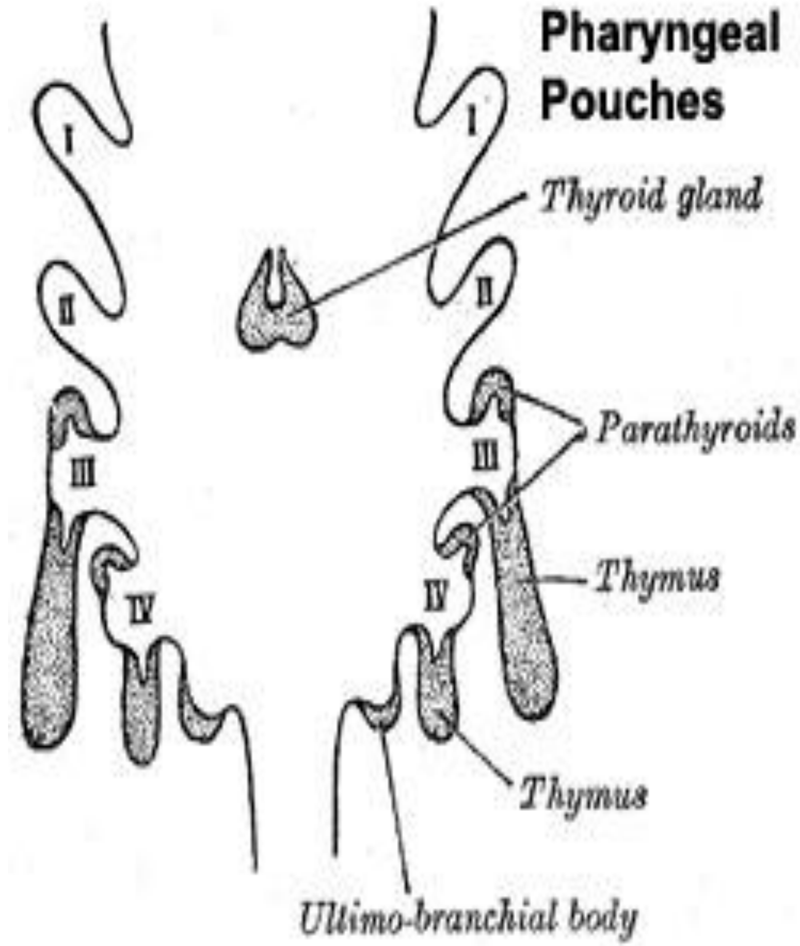
THIRD POUCH

The third pouch possesses Dorsal and Ventral wings.

Derivatives of the dorsal wings include the inferior parathyroid glands,

The ventral wings fuse to form the cytoreticular cells of the thymus

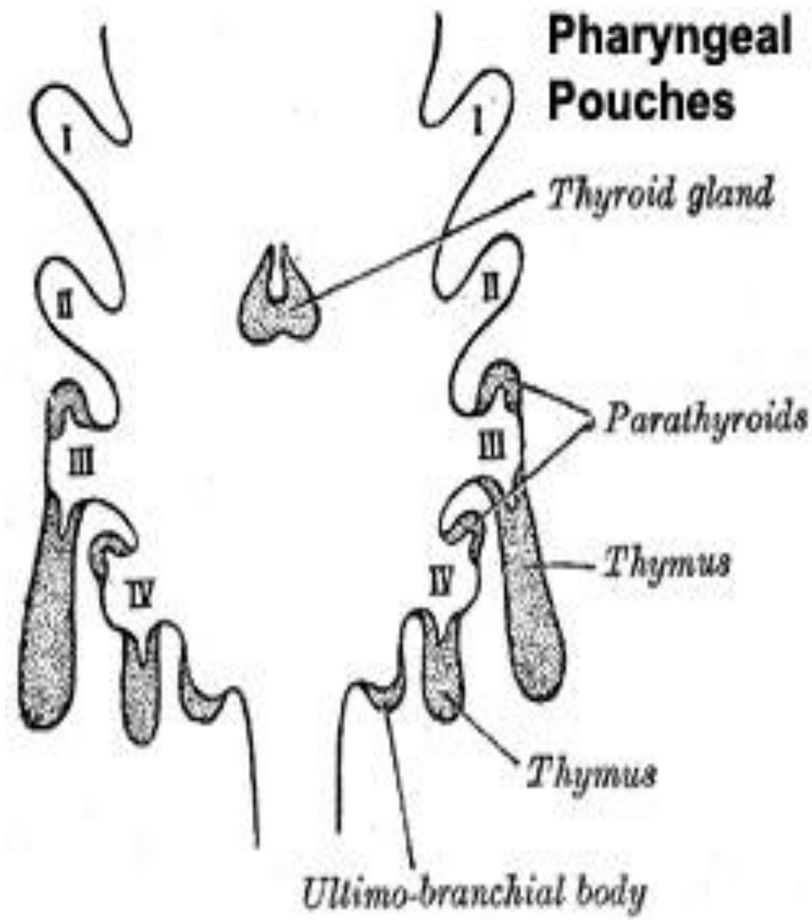
The main nerve supply to the derivatives of this pouch is Cranial Nerve IX, glossopharyngeal nerve.



FOURTH POUCH

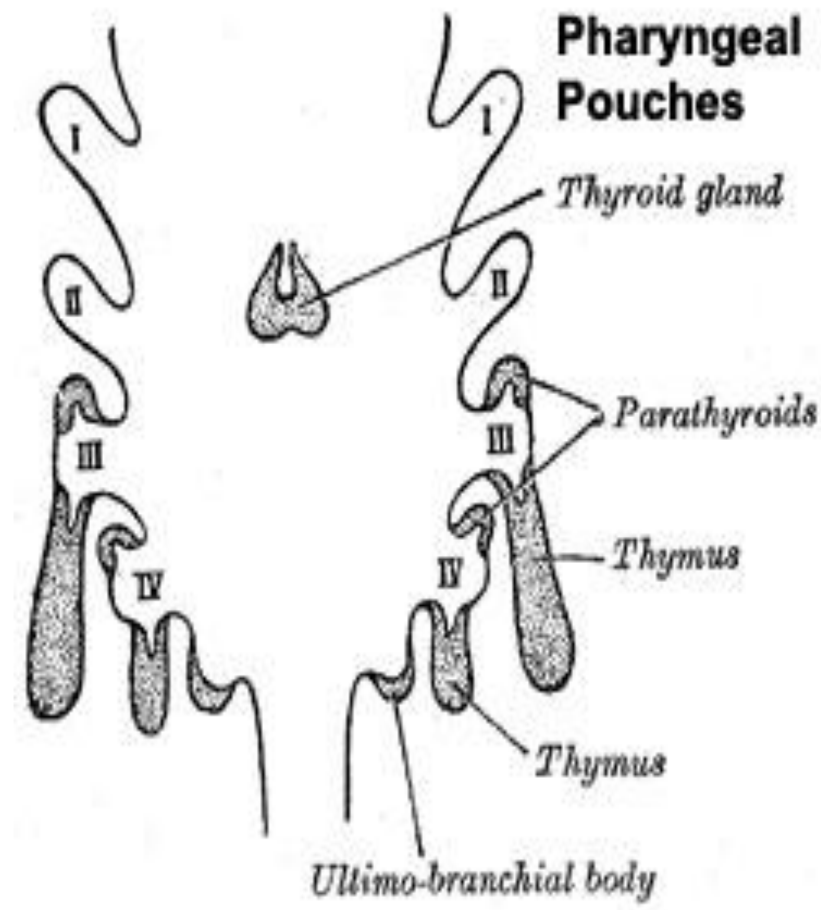
Derivatives include

- ***the superior parathyroid glands.***
- ***ultimobranchial body which forms the parafollicular C-Cells of the thyroid gland.***

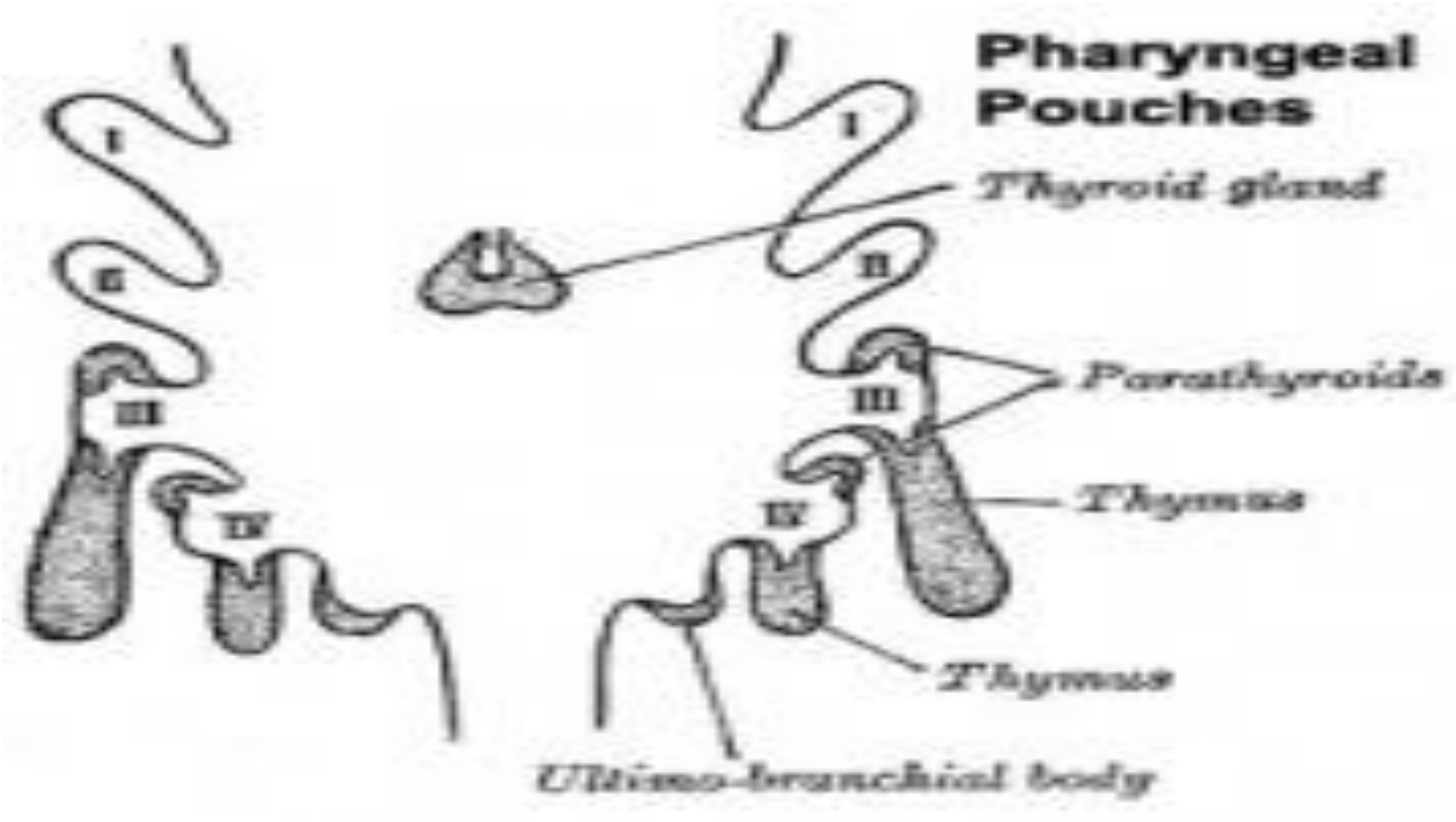


FIFTH POUCH

- ***Rudimentary structure, becomes part of the fourth pouch contributing to thyroid C-cells.***



PHARYNGEAL POUCHES

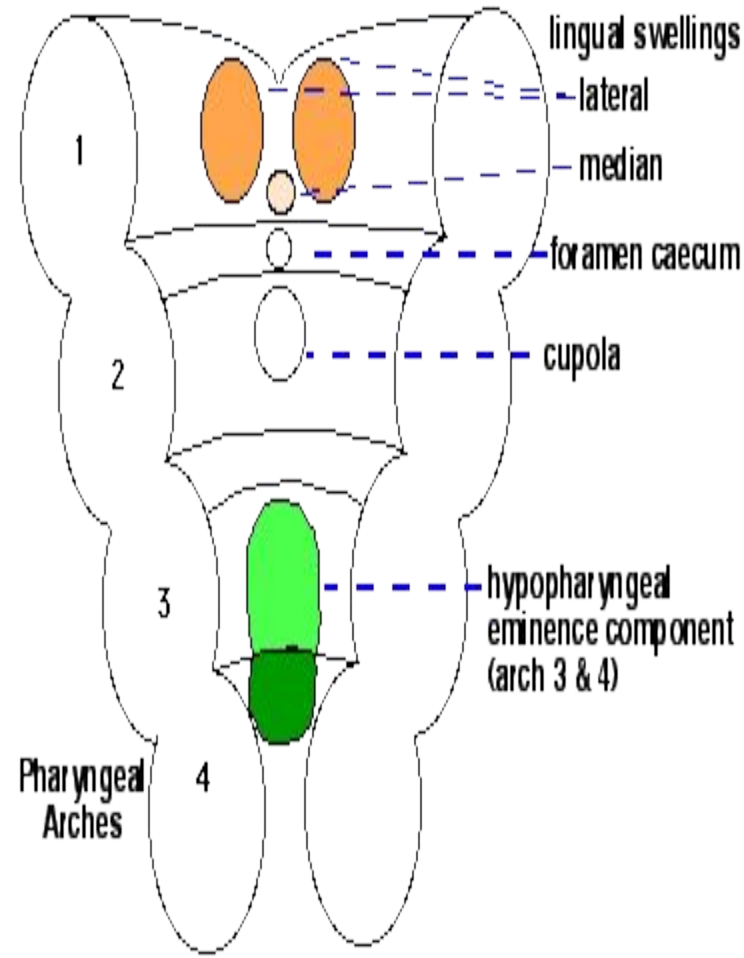


TONGUE

- ***The tongue begins to form at approximately the same time as the palates.***
- ***It extends from various protuberances on the pharynx floor.***
- ***Already at the time of the medial fusion of the first (mandibular) and second (hyoid) pharyngeal arches a medial protuberance, the tuberculum impar, appears on the lower edge of the mandibular arch.***
- ***To the left and right of it two further swellings form, the lateral lingual prominences.***

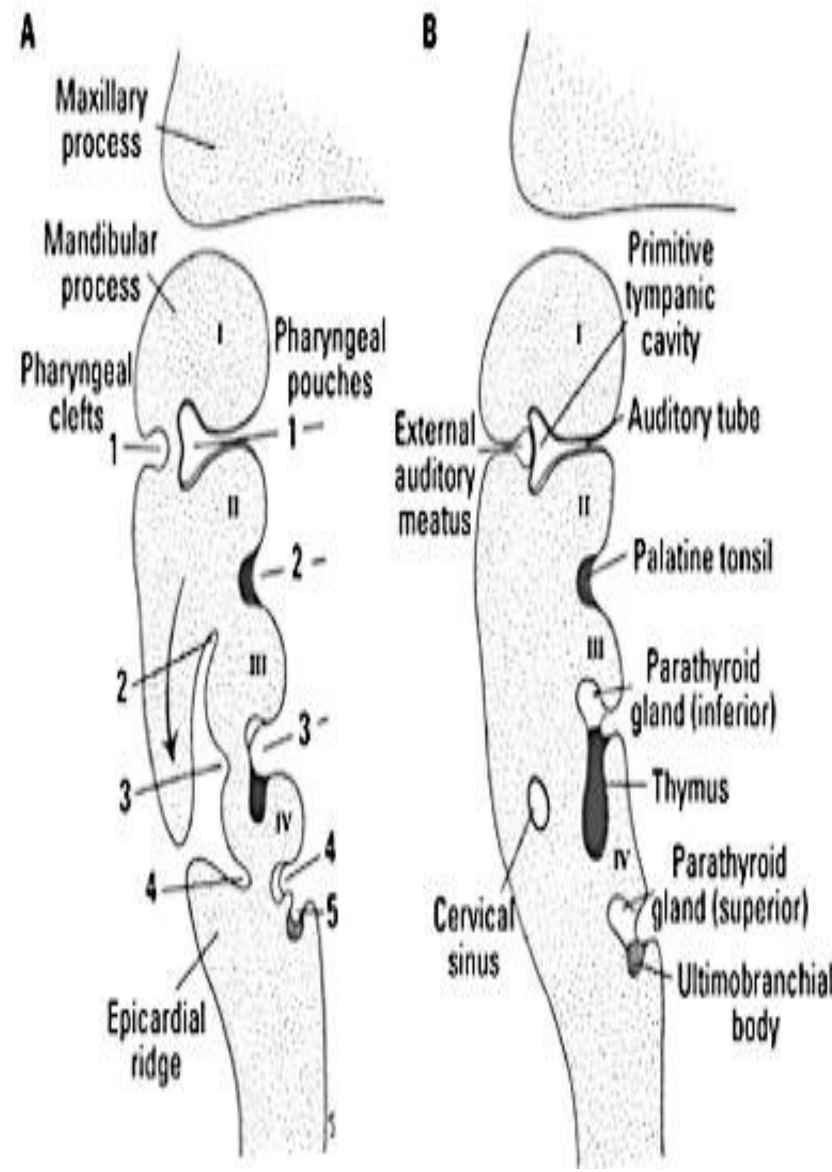
THYROID

- ***The primordium of the thyroid forms as a median thickening on the floor between the first and second pharyngeal arches.***
- ***Soon a bilobular sacculation of the endoderm develops out of it that remains connected with the mouth floor by the thyreoglossal duct.***
- ***This also gets thinner and thinner and, with the further descent, the epithelial cord disintegrates and the thyroid loses its contact with the mouth floor.***



THYMUS

- *The two main components of the thymus, the lymphoid thymocytes and the thymic epithelial cells, have distinct developmental origins.*
- *The thymic epithelium is the first to develop, and appears in the form of two flask-shape endodermal diverticula.*
- *Which arise, one on either side, from the third branchial pouch (pharyngeal pouch).*
- *Extend lateralward and backward into the surrounding mesoderm and neural crest-derived mesenchyme in front of the ventral aorta.*
- *They meet and become joined to one another by connective tissue, but there is never any fusion of the thymus tissue proper.*



THYMUS

- ***The pharyngeal opening of each diverticulum is soon obliterated, but the neck of the flask persists for some time as a cellular cord.***
- ***By further proliferation of the cells lining the flask, buds of cells are formed, which become surrounded and isolated by the invading mesoderm.***
- ***Additional portions of thymus tissue are sometimes developed from the fourth branchial pouches***